Issue date: 10/31/2003
Revision date: 8/31/2021

Safety Data Sheet

1. Chemical product and company identification

Product name : N,N-Diethyl-N'-1-naphthylethylenediamine oxalate

Company information

Name of manufacturer : KANTO CHEMICAL CO., INC.
Address : 2-1, Nihonbashi, Muromachi 2-Chome, Chuo-Ku, Tokyo, 103-0022, JP
Name of section : Business Administration Department, Reagent Division
Telephone number : +81-3-6214-1090
Facsimile number : +81-3-3241-1047
Mail address : BC32@kanto.co.jp
Reference No : 28086

2. Hazards identification

GHS classification

Health hazards	Skin corrosion/irritation	Category 2
	Serious eye damage/eye irritation	Category 2A

Hazard pictograms



Signal word : Warning

Hazard statements : Causes skin irritation
Causes serious eye irritation

Precautionary statements

Prevention : Wash hands, forearms and face thoroughly after handling.
Wear protective gloves/protective clothing/eye protection/face protection.

Response : IF ON SKIN: Wash with plenty of water.
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
Specific treatment (see supplemental first aid instruction on this label).
If skin irritation occurs: Get medical advice/attention.
If eye irritation persists: Get medical advice/attention.
Take off contaminated clothing and wash it before reuse.

3. Composition/information on ingredients

Distinction of substance or mixture : Substance

Synonyms : N-(2-Diethylaminoethyl)-1-naphthylamine oxalate, N-(1-Naphthyl)-N'-(diethyl)ethylenediamine oxalate, Tsuda's reagent

Chemical name	Concentration (%)	Formula	TSCA	EC-No.	CAS RN
N,N-Diethyl-N'-1-naphthylethylenediamine oxalate	≥ 98	C16H22N2・C2H2O4	Not listed	-	29473-53-8

4. First aid measures

First aid measures

First-aid measures after inhalation : Remove the victim to fresh air, and make him blow his nose and gargle.

First-aid measures after skin contact : Wash the affected areas under running water.

First-aid measures after eye contact : Wash the affected areas under running water.

First-aid measures after ingestion : Give the victim water or salt water and make him vomit. Get medical attention.

Personal Protection in First Aid and Measures : Rescuers should wear proper protective equipment like rubber gloves, goggles.

5. Fire fighting measures

Suitable extinguishing media : Water, dry chemical powder, carbon dioxide, dry sand, foam

Unsuitable extinguishing media : None

Firefighting instructions : Move containers from fire area if it can be done without risk, if not possible, apply water from a safe distance to cool and protect surrounding area.

Dry chemical powder, carbon dioxide or dry sand should be used for small fires. Foam extinguisher is effective for a large scale fire.

Personal protection (Emergency response) : Firefighters should wear protective equipment.

6. Accidental release measures

Personal Precautions, Protective Equipment and Emergency Procedures

General measures : Wear proper protective equipment and avoid contact with skin and inhalation of dust. Conduct operations from upwind and evacuate people downwind.

Environmental precautions

Environmental precautions : Attention should be given to avoid discharge of spilled product into rivers and resulting environmental damage. When diluting spill with large amounts of water, discharge of untreated wastewater into the environment must be avoided.

Methods and Equipment for Containment and Cleaning up

For containment : Sweep up in a chemical waste container. Flush contaminated area with copious amounts of water.

7. Handling and storage

Handling

Technical measures : Wear appropriate protective equipment to avoid contact with skin

or inhalation of dust.

Precautions for safe handling : The substance is an oxdizer. Avoid contact with organic substances.

Storage

Storage conditions : Store in a dark, cool place and tightly closed.

Material used in packaging/containers : Glass, polyethylene, polypropylene.

8. Exposure controls / Personal protection equipment

ACGIH TWA	Not established

Appropriate engineering controls : Install a local ventilation system in case of dusty condition.

Protective equipment

Respiratory protection : If necessary, wear dust mask

Hand protection : Impervious protective gloves

Eye protection : Safety goggles

Skin and body protection : Protective clothing, protective boots

9. Physical and chemical properties

Physical state : Solid

Color : White

Odor : Odorless

pH : No data available

Melting point : 167 ° C (Decomposition)

Freezing point : No data available

Boiling point : No data available

Flash point : No data available

Auto-ignition temperature : No data available

Decomposition temperature : No data available

Flammability (solid, gas) : Flammable solid

Vapor pressure : No data available

Relative density : No data available

Density : No data available

Relative gas density : No data available

Solubility : Water; Soluble. Organic solvents; Slightly soluble in ethanol.

Partition coefficient n-octanol/water (log Pow) : No data available

Explosive limits (vol %) : No data available

Viscosity, kinematic: : No data available

Particle characteristics : No data available

10. Stability and reactivity

Reactivity : Coupling with an aromatic diazonium salt to produce an azo dye.

Chemical stability : Stable under normal conditions. Colored with light.

Possibility of hazardous reactions	:	May react violently when in contact with oxidaizing substances.
Conditions to avoid	:	Light, heat.
Incompatible materials	:	Oxidizing substances.
Hazardous decomposition products	:	Carbon monoxide, nitrogen oxides.

11. Toxicological information

Acute toxicity (oral)	:	Classification not possible
Acute toxicity (dermal)	:	Classification not possible
Acute toxicity (inhalation)	:	No classification (gas) Classification not possible (vapor) Classification not possible (dust, mist)
Skin corrosion/irritation	:	Causes skin irritation Since the substance causes skin irritation, it was classified into category 2.
Serious eye damage/irritation	:	Causes serious eye irritation Since the substance causes eye irritation, it was classified into category 2A.
Respiratory sensitization	:	Classification not possible
Skin sensitization	:	Classification not possible
Germ cell mutagenicity	:	Classification not possible
Carcinogenicity	:	Classification not possible
Reproductive toxicity	:	Classification not possible There is no information about this substance. Oxalic acid is classified into category 2. Oxalic acid : In an 18-week continuous breeding study using mice administered by the oral route, it is reported that, at the high dose of 2000 ppm group, an increase in kidney weight and in abnormal sperm count was observed in F1 parent animals, and a decrease in the number of live fetuses and live female pups was observed in F2 animals. Therefore, this substance was classified into category 2.
STOT-single exposure	:	Classification not possible There is no information about this substance. In addition, respiratory irritant may be indicated according to oxalic acid. Oxalic acid : This substance binds to calcium in the blood, leading to hypocalcemia, affecting the nervous system. In humans, signs and symptoms including burning pain in the throat and stomach; esophageal erosion; nausea; vomiting; severe hypotension; weak and irregular pulse; headaches; weakness; convulsions; coma; and death by accidental or intentional ingestion of this substance are reported. Also, it is described that irritation of the respiratory tract and ulceration of mucous membranes are produced by inhalation of this substance. From the above, this substance was classified into category 1 (nervous system) and category 3 (respiratory tract irritation).
STOT-repeated exposure	:	Classification not possible There is no information about this substance. In addition, according to oxalic acid, affect may be indicated in the kidney. Oxalic acid : May affect the urinary system.As for humans, in Norwegian railroad workers (393, male) who responded in questionnaires regarding the prevalence of urinary stones, the reports of urinary stone colic increased depending on the categorization of exposure concentrations. Therefore, this substance was classified into category 1 (urinary organs).

Aspiration hazard	:	Classification not possible

12. Ecological information

Ecotoxicity

Aquatic acute	:	Classification not possible
Aquatic chronic	:	Classification not possible

Persistence and degradability

No additional information available

Bioaccumulative potential

No additional information available

Mobility in soil

No additional information available

Hazardous to the ozone layer

Ozone	:	Classification not possible

13. Disposal considerations

Ecology - waste materials	:	Mixed with flammable organic solvents and burn in a chemical incinerator equipped with an afterburner and a scrubber. Or entrust approved waste disposal companies with the disposal.
Contaminated container and packaging	:	In case of disposal of empty bottles, dispose bottles after removing the content thoroughly.

14. Transport information

International Regulations

Transport by sea(IMDG)

UN-No. (IMDG)	:	Not applicable
Proper Shipping Name (IMDG)	:	Not applicable
Packing group (IMDG)	:	Not applicable
Transport hazard class(es) (IMDG)	:	Not applicable

Air transport(IATA)

UN-No. (IATA)	:	Not applicable
Proper Shipping Name (IATA)	:	Not applicable
Packing group (IATA)	:	Not applicable
Transport hazard class(es) (IATA)	:	Not applicable
Marine pollutant	:	Not applicable

15. Regulatory information

Regulatory information with regard to this substance in your country or region should be examined by your own responsibility.

16. Other information

Data sources	:	Dictionary of Organic Compounds, The society of Synthetic Organic Chemistry, Kodansha Ltd.(1985). NITE Chemical Risk Information Platform (NITE-CHRIP), National

Institute of Technology and Evaluation.

The information contained herein is based on several references and the present state of our knowledge. However the SDS does not always cover all information about the product, handle the product carefully.The information is intended to ordinary usage, in case of particular handlings, conduct appropriate safety measurements.The information herein is only provision of information, and it does not represent a guarantee the properties of the product.The Safety Data Sheet(SDS) is prepared based on JIS Z7253.